

Yuma Oda

Research Engineer | ML Systems, Compilers, Inference

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🌐 github.com/hikettei

Research Engineer at Mirai Labs and second-year undergraduate at Keio University. My work spans ML systems, compilers, and efficient LLM inference, with a focus on speculative decoding for Apple devices. Numerical computing enjoyer and Common Lisp enthusiast.

EDUCATION

Keio University | *B.A. Candidate, Environment and Information Studies*

Apr 2025 - Apr 2029 (Expected)

Focus: Computer Science, Transactions, and Databases

EXPERIENCE

Mirai Labs | *Research Engineer*

Mar 2026 - Present

Full-time · Remote · San Francisco, CA, USA

- Co-developed **DFlash-TfM**, a state-of-the-art speculative decoding method, achieving a $4.37\times$ speedup over autoregressive decoding and 24.7% higher throughput than optimized DFlash on Qwen3.6-27B (BF16, single NVIDIA B200).
- Co-developed **speculative decoding in Uzu** for Apple M5-series Macs, achieving nearly $2\times$ the decoding throughput of MTPLX and over $3\times$ that of llama.cpp on Qwen3.6-27B at comparable quantization levels.

Aoha, Inc. | *Software Engineer*

Dec 2023 - Aug 2026

Part-time · Hybrid · Minato-ku, Tokyo, Japan

- Built deep-learning compiler infrastructure and neural network quantization methods for computer vision models.
- Researched efficient training methods for mixture-of-experts models.
- Built reinforcement learning infrastructure across 56 NVIDIA H200 GPUs for automated RTL design, including agent architecture, training harnesses, and data/evaluation pipelines.

PUBLICATION

Yuma Oda, Ryan Mathieu, Roman Knyazhitskiy, and Artur Chakhvadze. “Trees from Marginals: Autoregressive drafting with factorized priors.” *arXiv preprint arXiv:2607.06763*, 2026.

PERSONAL PROJECT & OPEN SOURCE CONTRIBUTION

Caten | *Creator and Maintainer*

- Created an experimental deep-learning compiler in Common Lisp using lightweight IRs, polyhedral compilation, and an optimizing pattern matcher.
- Designed a RISC-style graph IR that lowers computation to 26 composable primitive operations.
- Built lazy evaluation, Clang JIT compilation, model inference demos, and experimental training and automatic differentiation support.

tinygrad | *Fast-math approximations, PR 5187*

- Completed an approximation bounty by implementing SIN, LOG2, and EXP2 approximations; merged into tinygrad.

HONORS & AWARDS

METI Minister's Award, U-22 Programming Contest · \$3,600 award
Sokan no Susume Scholarship, Keio University · \$15,000 scholarship

Nov 2023

Apr 2025 - Apr 2029 (Expected)

LANGUAGES

Japanese: Native

English: Upper-intermediate (CEFR B2)